

1 Requirement crosswalk 1 of 3

The full stable prefix for each row is `SCI-FRUIT-REQ-`. Engineering interpretation adds no independent science.

Requirement	Scientist-facing source	Engineering interpretation
001	Section 8: A/B/C authorities	Keep program order, conditional composition and numerical method approval separate; retain numerical unavailability and a separate owner freeze decision.
002	Section 9: Scientific type identity	Bind quantity, unit/scale, frame/domain, grid/sampling, axes, timing, grouping, validity, causes and generations; shape or labels never substitute.
003	Section 9: Three model spaces	Bind candidate, accepted and applied objects/spaces and every transformation; equality of domain, support, unit, normalization, sign, representation, response, uncertainty or parentage needs an exact method/state fact.
004	Section 9: Pairwise additive compatibility	For removal and rejoin separately require exact quantity, unit/scale, frame/domain, grid/sampling, support, reference/gauge, null-space, calibration and response compatibility and declared affine/quotient arithmetic.
005	Section 10: Declared composition	Use the applied operand, typed maps, exact role order/bypass locations and counts; do not infer inverse, equal domains/support, upstream bypass or numerical defaults.
006	Section 10: Linearity hierarchy	Use the map-valued expansion only for exact fixed linear F; define D and its linear model response/covariance only when A, Pi and B are all exact fixed linear.
007	Section 15: Parent and input rule	Retain immutable measured-base ancestry and the exact method-defined iteration input with required domain conversions; returning to base differs from prior-result processing.
008	Section 11: Eight support roles	Retain all eight support/influence roles, their exact queries and path/application identities; a numerical method cannot waive a missing role.

Admitted source: All eleven admitted payloads; scoped details and source attributions in the cited scientific sections.

2 Requirement crosswalk 2 of 3

The full stable prefix for each row is `SCI-FRUIT-REQ-`. Engineering interpretation adds no independent science.

Requirement	Scientist-facing source	Engineering interpretation
009	Section 11: Occurrences and mode origin	Use only permitted residual-and-model, residual-only, model-only or unavailable occurrence classes; retain model/prior origin in unavailable modes with no measurement/exposure/recovery inference.
010	Section 15: Method versus state	Rule changes require distinct method identities; new realizations under unchanged rules have distinct state/iteration/application generations.
011	Section 15: Operation-state factorization	Assign exactly one of seven categories per factored operation/state component at each conditioning scope, with owner, parents, dependencies, order, count, path and failure.
012	Section 12: Response identities	Bind all required RF-01 through RF-07 identities and statuses with exact query convention, domains, fixed/rerun facts, scope, comparability and unavailable/discontinuity behavior.
013	Section 13: Response composition	RF-03 is local with prior state fixed; RF-04 fixes extent; RF-05 includes stop/resource/selector; RF-07 composes the parent-domain query using components; RF-06 reruns included upstream producers.
014	Section 14: Complete covariance	Use exact applied-model joint conditioning and all four covariance terms under complete linearity/finite-second-moment premises; transform candidate/accepted covariance and preserve cross dependence.
015	Section 14: Uncertainty target	Separate nine uncertainty roles, omitted variation and reference/gauge/null-space consequences; fixed gauge is not measured zero uncertainty; repeated iterations are not independent measurements.
016	Section 14: NOI distinctions	Require exact fixed-state application parity; identify dependent successor generations; separate fixed-state/relearned method and ensemble rules without pooling, retaining member-specific states.

Admitted source: All eleven admitted payloads; scoped details and source attributions in the cited scientific sections.

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Requirement	Scientist-facing source	Engineering interpretation
017	Section 16: Continuation sufficiency	After absolute zero-based index N continue at N+1 from sufficient S_{N+1} , with the same future controls/profile; preserve all causal queries, reconstruction and failure contracts.
018	Section 16: Lifecycle identity	Keep seed, continuation, restart, relearned successor, branch, retry, completion, selection and publication distinct; restart creates a new lineage.
019	Section 16: Termination propositions	Separate completion/resource/convergence/policy-stop/failure/selection/fitness; bind convergence proposition, metric, scale, tolerance, direction, persistence, causes and generation without selecting values.
020	Section 16: Immutable terminal record	Require all nine terminal fields, including full candidate population and response/uncertainty; missing hard selection dependencies block the claim; never mutate/relabel/replace the selected bundle.
021	Section 17: Universal completion	Preserve every universal bundle role and hard result/continuation/response/uncertainty/selection dependency; narrower unavailable claims require both core and exact method permission.
022	Section 17: Owner boundaries and grouping	Retain producer science and downstream admission; one complete grouping is method identity; no coadd commutation, generic new parent class or SCI-VAL execution of recurrence/selection is implied.
023	Section 8: Complete method and gate	Require every scientific method-record field, its exact method-specific identity/owner decision and upstream numerical authority before any numerical route/result; a completed template is not approval.
024	Section 17: Predictions and nonclaims	Preserve SCI-FRUIT-PRED-001 through 020 with explicit premises and unavailable claims; document verification establishes no numerical qualification or downstream fitness.

Admitted source: All eleven admitted payloads; scoped details and source attributions in the cited scientific sections.

4 Required numerical-method record 1 of 3

FRUIT-FEEDBACK-METHOD v0.1 is a required-record family, not an approved numerical instance. Every row below is required. Actual scientific values remain `unavailable_pending_separate_owner_approval`.

Authority and lineage

Method-specific identity/revision under the family, exact owner decision and authority state; method generation distinct from state/iteration/application generations; immutable base parent and producer lineage

Parent route and grouping

One exact numerically available route supplying quantity, unit, frame, response, beam/template convention, calibration, support, validity, lifecycle and provenance; one declared grouping with membership, axes and scope; all required domain conversions

Iteration-input rule

Immutable original parent, previous z_{k-1} , reconstruction from base parent/current cumulative model, or another exact product; every conversion and its authority

Target and model

Separate $M_k^{\text{candidate}}$, M_k^{accepted} , M_k^{applied} and their model objects; target/model quantities, distinct external versus internally learned origin and prior, and all transformations, if any; no assumed equality of domain, support, unit, normalization, sign, representation, response, uncertainty or parentage; equality requires an exact method/state fact

Candidate/selection/acceptance/application

Exact construction, causal selection/acceptance and acceptance-to-application rule; allowed inputs at action time; every normalization, clipping, sign conversion, support restriction, projection, unit conversion or other transformation; equality of accepted and applied objects only by exact method/state binding

Model accumulation meaning

Cumulative model versus increment versus diagnostic difference; exact relation to the separate accepted/applied models and successor state

Typed composition

Y_k , $M_k^{\text{candidate}}$, M_k^{accepted} , $M_k = M_k^{\text{applied}}$, Z_k ; Π_k , F_k , B_k and domains/codomains, units, support, state and conditioning; applied model as operand; sign convention and exact removal/rejoin locations; each linearity claim explicit

Admitted source: FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD (complete binding table, exact scientific obligations).

5 Required numerical-method record 2 of 3

FRUIT-FEEDBACK-METHOD v0.1 is a required-record family, not an approved numerical instance. Every row below is required. Actual scientific values remain `unavailable_pending_separate_owner_approval`.

Additive compatibility and unavailable modes

Declare exact additive Y_k/Z_k or explicit affine/quotient spaces with one reference/gauge for each operation and well-defined arithmetic. For both removal and rejoin bind scientific quantity, unit/numerical scale, frame/domain, grid/sampling, support, additive origin/reference or gauge, null-space state, calibration and response conventions; exact alignments/conversions, unavailable-mode origin and failure consequences

Operation graph

Ordered named operations, exact bypass set, all application counts and parent/state bindings; no generic bypass flag or duplicated response application

State by operation

Exactly one category per named operation and consequential state component from the operation-state table; every fixed, recomputed, externally supplied or relearned fact, with learning schedule and triggering conditions

Support and missing data

All path supports, learning influence, response-query support, information-origin classes, unequal-support rules and failure scope

Update and transition

Exact rule consuming S_k and y_k^{in} and producing separate $m_k^{\text{candidate}}$, m_k^{accepted} , m_k^{applied} , ρ_k , z_k and S_{k+1} , with core-permitted typed not_applicable/unavailable roles; no default recurrence

Response

All response-role identities and typed statuses required by the applicable method and claim, from the exact seven-role catalog; perturbation source/domain, fixed/rerun state, recurrence/terminal scope, baseline/perturbed comparability, branch/support/state changes, codomain, derivative or finite-difference convention, unavailable/discontinuity behavior; additive reference/gauge and null-space consequences

Uncertainty

Exact joint conditioning, covariance or NOI method; applied-model covariance and both measured-input/applied-model cross terms under complete linearity; transformations, reference/gauge/null-space conditioning and model-supported modes, omitted sources, ensemble and state generations; unavailable modes are not zero uncertainty by implication

Admitted source: FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD (complete binding table, exact scientific obligations).

6 Required numerical-method record 3 of 3

FRUIT-FEEDBACK-METHOD v0.1 is a required-record family, not an approved numerical instance. Every row below is required. Actual scientific values remain `unavailable_pending_separate_owner_approval`.

Continuation sufficiency

Complete continuation-state query vocabulary; same-future-controls equality criterion under the declared numerical profile; branch/generation compatibility

Stochastic state

Exact generator/algorithm, keys/counters/state, populations and sampling controls where applicable; otherwise justified not_applicable

Convergence proposition

Exact proposition, quantity/domain, metric or norm, scale/denominator, tolerance, direction, persistence/consecutive rule, missing/non-finite behavior, generation and state

Resource limits

Exact accounting, counters, stopping consequences and exhaustion state; no default cap or inference of convergence

Terminal selection

One exact predeclared rule, generation, data access, complete candidate iteration population and ties/failure behavior; record selected completed-iteration identity, convergence/stopping/resource-limit facts, selection causes, selected-terminal response/uncertainty, continuation or terminality state and downstream named-use status references; see required terminal record in lifecycle table

Persistence/reconstruction

Materialized, compact or lineage-reconstructible representation; exact state queries and path components preserved; retention/reconstruction and failure contract

Completion, failure and publication

Universal core minimum bundle and hard claim dependencies, additional method-required roles and named-use optional companions; permitted narrower claims with truthful unavailable status; exact failure actions/scope, immutable terminal selection and publication status; no waiver of core requirements

Admitted source: FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD (complete binding table, exact scientific obligations).

7 Decision authority and the next owner boundary

A. Program order: approved

SCI-FRUIT-OD-PROGRAM-ORDER-2026-09-06 permits conditional authorship before numerical approval/qualification. It selects no numerical sequence.

B. Conditional composition: approved

SCI-FRUIT-OD-CONDITIONAL-COMPOSITION-2026-09-06 permits the typed remove/process/rejoin consequences under explicit premises. It selects no maps, recurrence, support, grouping or state realization.

C. Numerical method: unavailable

FRUIT-FEEDBACK-METHOD v0.1 is the required-record family. Every numerical instance needs its own exact FRUIT-FEEDBACK-METHOD/<method-name>@<revision> identity, complete record, separate Grant Wilson approval and required upstream authority. Availability remains `unavailable_pending_separate_owner_approval`. A or B does not imply C.

The approval/launch control SCI-FRUIT-OD-STAGE-A-R0.4-AMENDED-APPROVAL-2026-09-07 closes ODQ-012 for the exact verified packet. It supersedes historical pending approval/pre-launch wording only, with no payload science change. The packet and approval control remain unchanged.

The owner ledger preserves ODQ-001 and ODQ-001A-001F, ODQ-002-011 as separate deferred numerical/producer matters. Their absence is intentionally compatible with generic conditional authorship and a later owner decision on conditional freeze. No unresolved additional scientific premise is invented or answered by this document. The author decision log records presentation and bounded logical derivations, not new owner approvals.

Current response supersession preserves RF-01; narrows RF-02 explicitly to applied model; binds RF-03 to one iteration with prior state fixed; makes RF-04 fixed-sequence/index; includes stop/resource/selection in RF-05; reruns included upstream in RF-06; and retires RF-07's held-numerical-parent meaning in favor of the complete composed FRUIT-only parent-domain query. RF-07 references RF-03/RF-04/RF-05 rather than duplicating them. Within FRUIT, earlier "exact restart" wording is superseded by same-lineage continuation; restart creates a new lineage. No upstream lifecycle is edited.

SOURCE_IDENTITIES.md, OWNER_LEDGER.md, DECISION_LOG.md and REQUIREMENT_CROSSWALK.md accompany this authority. The review report records independent consistency findings within the three-substantive-round limit; DOCUMENT_VERIFICATION.md records only artifact checks. Neither review nor compilation freezes the core, admits a numerical method or supplies validation. Numerical development, qualification and any policy recommendation retain their separate owner decisions and independent-pointing evidence obligations. The evidence plan is known-but-unsupplied here and is not invented. The next decision is Grant Wilson's separate owner review/conditional freeze.

Admitted source: SCIENTIFIC_OWNER_DECISION_LEDGER; PROGRAM_REFERENCE; approval addendum (launch control only).

8 What protecting a model can mean

SCI-FRUIT: conditional feedback core

Contract v0.1 Document r0.1 7 September 2026

Stage B draft for owner review; not frozen

Input. Exact iteration input y_k^{in} , immutable measured-parent ancestry, scientific state S_k , separately typed candidate/accepted/applied model roles, and a separately approved method if numerical use is ever sought.

Output. Conditional meaning of an immutable completed-iteration bundle, or a separate terminal-selection record plus its selected bundle.

Equation. On compatible declared spaces and support, $\rho_k = y_k^{\text{in}} - \Pi_k(m_k^{\text{applied}})$ and $z_k = F_k(\rho_k) + B_k(m_k^{\text{applied}})$.

Sources. Only the eleven verified author payloads of Stage A r0.4 amended; SOURCE_IDENTITIES.md binds their exact bytes.

Status. Generic conditional authorship authorized. Every numerical method and route: `unavailable_pending_separate_owner_approval`. Implementation unassessed; numerical validation unperformed; separate owner review/conditional freeze pending.

Removing a declared model contribution before a residual operation, then rejoining its declared contribution, can protect that model path from the *exactly named* residual operation. It does not establish that the model is correct, that its content is true sky, or that a mode missing from the measured path has been observed. The useful scientific question is therefore which quantity traveled along each path, with what reference, support, dependence and response, and what claim the combined result can sustain.

The boundary begins at the exact iteration input and state, retaining the original measured parent. It includes model construction, selection, acceptance and application, removal, residual processing, bypass, rejoin, successor state and iteration/terminal identity. It ends at the bundle and selection record. Downstream inference and use retain separate authorities. No recurrence, numerical operator or stopping policy is implied by this story.

Program adherence and prior-work recovery

The admitted PROGRAM_REFERENCE supplies the program and reuse obligations. Stage A recovery is retained by identity through this approved packet; it was not reopened. The permitted adjacent extracts are reused under their covers, with only the scoped FRUIT terminology supersession recorded in the ledger. The frozen PTC fragment is preserved exactly. No unlisted source, code, configuration, prior conversation, reduction or empirical product enters this authorship. Local compilation and inspection supply document evidence only.

Authority A permits this authorship order; authority B permits the conditional composition; authority C requires a separately approved exact numerical method. A or B never implies C. Packet approval closes ODD-012 only. Approved facts, known-but-unsupplied producer/method facts, open scientific decisions, unassessed implementation and unperformed validation remain distinct.

Admitted source: PROGRAM_REFERENCE; SCOPE_BRIEF §§1–2,8,10–11; SCIENTIFIC_OWNER_DECISION_LEDGER.

9 The model applied to the data is a separate object

A model accepted for feedback need not be the operand removed from the input. Acceptance may precede normalization, clipping, sign conversion, support restriction, projection, unit conversion or another declared transformation. Scientific accountability therefore needs three objects, even when their numerical values happen to agree:

$$m_k^{\text{candidate}} \in M_k^{\text{candidate}}, \quad m_k^{\text{accepted}} \in M_k^{\text{accepted}}, \quad m_k^{\text{applied}} \in M_k = M_k^{\text{applied}}.$$

The first is a proposed external or internally constructed candidate; the second is the model accepted for feedback; the third is the exact operand used at removal and rejoin. Bind every transformation, if any, among these spaces and objects, with its rule, causal inputs, state and generation. Equality of any pair requires an exact method/state fact. Do not assume equal domain, support, unit, normalization, sign, representation, response, uncertainty or parentage. A covariance or perturbation attached to the accepted model is not automatically a covariance or perturbation of the applied model.

The input and result have their own spaces, $y_k^{\text{in}} \in Y_k$ and $z_k \in Z_k$. The applied-model space need not equal either. Each space binds physical quantity, unit and numerical scale, coordinate frame/domain, grid/sampling, shape and axis/index meaning, timing where relevant, grouping, support, validity, missing/non-finite causes and generations. The declared target is also separate: neither the measured input nor an accepted model is true sky by identity. External models retain their external parent and prior; internally learned models retain learning parents/population, selection, response, uncertainty and state generations. Their claim here is only **feedback_model_state**.

Subtraction requires a common physical reference

Y_k and Z_k are exact additive quantity spaces, or explicitly identified affine/quotient spaces with one declared reference and gauge for each operation and well-defined arithmetic. An affine method must supply exact additive coordinates; a quotient method must supply well-defined quotient arithmetic. Arbitrary arithmetic on representatives is not licensed.

For the removal pair and, separately, the rejoin pair, bind compatible *scientific quantity; unit and numerical scale; coordinate frame/domain; grid/sampling; support; additive origin/reference or gauge; null-space state; calibration convention; and response convention*. Record every conversion or reference/gauge alignment explicitly. Agreement within a pair does not imply $Y_k = Z_k$. Equal shape, unit labels or masks do not establish compatibility. A missing compatibility premise makes the affected arithmetic unavailable.

An unavailable absolute or other additive mode remains observationally unavailable after rejoin. Choosing a gauge changes no measurement authority. Carry the exact reference, gauge, null-space and mode-origin consequences into support, response, uncertainty and the bundle.

Admitted source: NOTATION_AND_ROLE_TAXONOMY; SCOPE_BRIEF §§2–3; BOUNDARIES_AND_CONVENTIONS B1.

10 Removal and rejoin determine the conditional result

Bind $\Pi_k : M_k \rightarrow Y_k$, $F_k : Y_k \rightarrow Z_k$ and $B_k : M_k \rightarrow Z_k$. $\Pi_k(m_k^{\text{applied}})$ is the declared removal contribution; $B_k(m_k^{\text{applied}})$ is the declared contribution at rejoin. Only when the two operand pairs have the compatibility in Section 9 does the ordinary relation apply:

$$\begin{aligned}\rho_k &= y_k^{\text{in}} - \Pi_k(m_k^{\text{applied}}), \\ z_k &= F_k(\rho_k) + B_k(m_k^{\text{applied}}).\end{aligned}\tag{1}$$

The model path bypasses the exactly named residual operation. The operation graph binds order, removal/rejoin locations, all path application counts and consequential state. It does not imply bypass of every upstream operation. There is no assumed $\Pi_k = B_k$, inverse relation, equal domain/support, linear F_k , true model or unit response. Missing path contributions are not zero and undefined partial-domain maps are not silently extended.

What follows when only the residual operation is linear

If the complete $F_k = A_k$ is exact fixed linear on the declared spaces, support, state and conditioning, distributivity gives

$$z_k = A_k y_k^{\text{in}} - A_k[\Pi_k(m_k^{\text{applied}})] + B_k(m_k^{\text{applied}}).\tag{2}$$

This remains a map-valued expression. Nonlinear Π_k or B_k prevents a linear applied-model operator from being inferred. A fixed loading basis alone is insufficient; coefficient recomputation alone neither proves nor disproves complete linearity. Centering, support and every material state belong to the claim. An affine offset cannot be discarded to obtain this form. Other procedures retain Equation 1 and their full state graph.

The stronger, completely fixed linear specialization

Only if A_k, Π_k, B_k are *all* exact fixed linear maps on the stated spaces, units, support, state and conditioning may one define

$$D_k := B_k - A_k \circ \Pi_k : M_k \rightarrow Z_k, \quad z_k = A_k y_k^{\text{in}} + D_k m_k^{\text{applied}}.\tag{3}$$

The difference operator is well typed because both terms map the applied-model space into one compatible result space. It describes the conditional net influence of applied feedback, not all sky bias. Holding the same input and operators and changing $m_k^{\text{applied}} = m_{\text{ref}}^{\text{applied}} + \delta_m^{\text{applied}}$ gives

$$z_k(m_k^{\text{applied}}) - z_k(m_{\text{ref}}^{\text{applied}}) = D_k \delta_m^{\text{applied}}.\tag{4}$$

Its conditional mean is the corresponding model-induced bias component. Accepted-model error requires its declared transformation first. Zero/identity limiting cases in the predictions are deductions under premises, not defaults or admitted run plans. None restores an upstream unavailable measured mode.

Admitted source: RESPONSE_AND_UNCERTAINTY_FAMILIES; SCOPE_BRIEF §2; REQUIRED_CONDITIONAL_PREDICTIONS.

11 Supported output is not the same as measured information

A finite output value says neither where its information came from nor where its defining operation is valid. Preserve eight separate roles:

1. **Measured-parent support:** original measured occurrences, producer availability, domain and causes.
2. **Model-definition support:** separately defined candidate, accepted and applied domains, origin and transformations.
3. **Projection/removal support:** where the projected applied model and compatible subtraction exist.
4. **Residual-processing support:** each operation's input/output support, retained membership and causes.
5. **Model-rejoin support:** where the rejoin contribution is defined, independently of removal.
6. **Final output support:** the method-defined output domain and permitted path composition.
7. **Learning influence:** transitive dependence of learned/selected state on inputs, not local membership or exposure.
8. **Response-query support:** exact family, source basis/domain, conditioning, output support and limits.

None can be waived. Materialize them or preserve exact reconstruction with path and operation/application identities. A common mask name is not identity. Acceptance does not imply application on the same support. Pairwise additive compatibility remains required in addition to support compatibility.

Four occurrence classes, with separate mode origin

`residual_and_model_supported` means both contributions exist and may be combined under the exact method. `residual_only` or `model_only` requires explicit method permission for that narrower path composition; absence of the other contribution is not numerical zero. `unavailable` retains the cause and affected domain when no permitted composition exists. A method may prohibit any class. Merely having a model at an unmeasured location does not permit model-only output.

Undefined unequal-support behavior makes the affected output unavailable. There is no implicit zero fill, extrapolation, imputation, renormalization, nearest substitution, offset/gauge alignment or completion of missing measured support. A method must bind any mapping or restriction into a common result domain.

An upstream unavailable absolute or other additive mode remains `model_or_prior_supported` if content is rejoined there. This is a mode-level annotation, *not a fifth occurrence class*: even a `residual-and-model-supported` occurrence may contain such a mode. It is not `observationally_measured`, `recovered_from_data`, `acquired_exposure` or `unit_response_truth`. A gauge creates no observation. Carry mode identity and reference/gauge consequences into response and uncertainty.

Iteration, projection and rejoin create no acquired exposure or independent observation. Shared parents, models, calibration, response and reference information preserve dependence. Transitive learning influence is not exposure; no producer exposure product or reconstruction is supplied without its authority.

Admitted source: `SUPPORT_INFORMATION_ORIGIN`; `NOTATION_AND_ROLE_TAXONOMY`.

12 Response depends on what is perturbed and what is held fixed

A response is an exact query about a procedure. Bind perturbation source/domain, direction/basis, units, limit or finite step and normalization when relevant; fixed versus rerun state; recurrence and terminal scope; baseline/perturbed comparability; branch/support/state changes; codomain; derivative or finite-difference convention; application counts, generations, conditioning; and unavailable/discontinuity behavior. Reference/gauge/null-space comparisons must be compatible. A changed reference is not silently a signal perturbation. No numerical query or step is selected here.

RF-01: fixed one-step measured-input response

Perturb y_k^{in} in Y_k , holding the applied model and declared maps, state and support fixed. Evaluate the unchanged one-step composition; no learning, recurrence, stopping or terminal selection. Codomain: Z_k differences. Measured-input names this path, not necessarily the original parent or an independent observation.

RF-02: fixed one-step applied-model response

Perturb m_k^{applied} in M_k^{applied} , holding the measured iteration input and declared maps, state and support fixed. Evaluate the unchanged one-step composition; no acceptance/model learning, recurrence or selection. Codomain: Z_k differences. Candidate or accepted perturbations require their transformations in the corresponding full procedure.

RF-03: one-iteration FRUIT full-procedure response

Perturb the declared one-iteration input on its exact domain. Hold prior S_k and upstream producer procedure/state fixed. Rerun construction, selection, state resolution, support, coefficient realization and application for iteration k only. Codomain: its Z_k response. No earlier recurrence or terminal selector belongs to this local role.

RF-04: fixed-sequence recursive response

Perturb the declared recursion input. Hold method rules, initial state, upstream producer procedure/state and the predeclared fixed sequence or absolute endpoint index fixed. Propagate input/model/state dependence through that extent, binding exact input and result/sequence codomains. Data-dependent sequence length and terminal selection are outside this role.

All roles retain branch/support/state changes and typed unavailability or transitions when comparisons/results/derivatives do not exist. Under complete fixed linearity in Equation 3, the separate one-step responses are

$$R_{z \leftarrow y, k}^{\text{FRUIT, fixed}} = A_k, \quad R_{z \leftarrow m^{\text{applied}}, k}^{\text{FRUIT, fixed}} = D_k. \quad (5)$$

If only $F_k = A_k$ is linear, the fixed input response may still be A_k ; the model response needs the derivative of the complete map-valued expression where it exists, not an inferred D_k .

Admitted source: RESPONSE_AND_UNCERTAINTY_FAMILIES; NOTATION_AND_ROLE_TAXONOMY.

13 Terminal, FRUIT-only and whole-chain queries have different reach

RF-05: selected-terminal response

Perturb the declared selection/procedure input and name all fixed initial and upstream facts. Include consequential iterations, convergence, stopping, resource limits and terminal selection, with complete candidate population and causes. Bind input and selected-result codomain, query convention and terminal record. Include branch/support/state/selection transitions and reference/gauge/ null-space comparison. Unavailable dependencies or nonexistent derivatives remain unavailable or typed transitions.

RF-06: whole-chain response

Perturb the exact upstream/source query on its domain, naming fixed external facts. Rerun every included data-dependent upstream producer and the declared FRUIT procedure, including its recurrence/terminal scope. Bind source and final domains and convention; each included producer and comparison must be available. Otherwise retain exact unavailability/discontinuity.

RF-07: complete FRUIT-only parent-domain response

Perturb the realized parent-domain payload while holding the upstream producer operation, learned state and generation fixed. Compose the parent-domain entry with the complete FRUIT procedure, including recurrence and terminal selection as applicable. Bind parent-query/final-response domains, initial FRUIT conditions, query convention and RF-03/RF-04/RF-05 component references. Include branch/support/state/selection and reference/gauge/null-space consequences. No upstream producer is rerun.

RF-03 supplies local iteration relations. RF-04 propagates fixed extent; RF-05 includes the selector and data-dependent stopping extent. RF-07 binds the outer realized-parent query to those components: it is not an independent alias of RF-03, RF-04 or RF-05. When a one-iteration special case reduces to a component, record that exact specialization and reuse its bound response/status. Equal values do not erase query identity. RF-06 additionally reruns included upstream producers. The previous RF-07 held-numerical-parent meaning is retired.

Perturbed parent payloads are separately identified response-query objects with immutable baseline-parent ancestry. A fixed producer generation does not mutate the measured parent or claim a new upstream realization. A rule-changing control perturbation requires a new method identity and explicit comparison of methods, not a state perturbation disguised as one rule.

For RF-01/RF-02, “fixed” means the conditioned map and unchanged application rule. Coefficients recomputed by that rule from a perturbed operand must still be recomputed, with new coefficient/application states. Freezing their numerical values instead changes the procedure. Missing baseline/perturbed codomain compatibility makes the response unavailable. Threshold, branch, support, state or selector changes may defeat derivative existence; a finite difference needs its own convention, never a substituted fixed-branch derivative.

Use all response identities/statuses required by the applicable method and claim. The seven-role catalog does not make any numerical response universally available. No online adaptive estimator is introduced by these names.

Admitted source: RESPONSE_AND_UNCERTAINTY_FAMILIES; SCIENTIFIC_OWNER_DECISION_LEDGER (response crosswalk).

14 Shared input and model information must remain in uncertainty

Assume complete fixed linearity, compatible real vector representations, finite joint second moments and exact conditioning Ω_k^{FRUIT} fixing relevant operators/state, reference/gauge/null-space coordinates and support. Define $C_{yy}^{\text{FRUIT}}, C_{mm}^{\text{FRUIT}}, C_{ym}^{\text{FRUIT}}, C_{my}^{\text{FRUIT}}$ as the joint conditional covariance blocks of $(y_k^{\text{in}}, m_k^{\text{applied}})$, with exact quantity axes, units, domains and conditioning. The m axis denotes only the applied model. Equation 3 then gives

$$\begin{aligned} C_z^{\text{FRUIT}} &= A_k C_{yy}^{\text{FRUIT}} A_k^T + D_k C_{mm}^{\text{FRUIT}} D_k^T \\ &\quad + A_k C_{ym}^{\text{FRUIT}} D_k^T + D_k C_{my}^{\text{FRUIT}} A_k^T. \end{aligned} \tag{6}$$

This follows by expanding the conditional covariance of the sum; the two mixed products are the two cross terms. In a valid real joint covariance, $C_{my}^{\text{FRUIT}} = (C_{ym}^{\text{FRUIT}})^T$. Do not drop either term without exact independence/conditioning authority establishing its absence. A model learned from the parent is not independent by implication. Candidate/accepted covariance needs the exact transformation and its dependence before it can supply applied covariance.

If the applied model is deterministic under the conditioning, its variance and cross blocks vanish there, leaving $A_k C_{yy}^{\text{FRUIT}} A_k^T$. This conditional target omits model-learning variation; it is not unconditional independence or covariance completeness and cannot erase unknown sources from another target. Random-operator/nonlinear procedures need their own uncertainty method. A Jacobian approximation is separately typed, not this exact identity.

Keep nine uncertainty roles distinct: measured-parent covariance; candidate/accepted/applied model/prior covariance; parent-model cross covariance; operator/state uncertainty; support/threshold selection; stopping/terminal selection; model mismatch/bias; empirical repeatability; and NOI uncertainty. Bias is not covariance by identity. Each binds reference/ gauge, unavailable modes, model/prior support, omitted mode/reference variation and exact conditioning. A fixed gauge or singular covariance does not mean an unavailable absolute mode was measured with zero uncertainty. Selected-terminal uncertainty additionally binds the full candidate population, stop/resource facts, selection record and causes. Reused parents imply no independent observations or $1/\sqrt{\text{iteration count}}$ uncertainty law.

Fixed-state NOI applies identical fixed FRUIT state/operator graph with exact application parity to every compatible admitted realization; absent parity makes the numerical route unavailable. It omits unmodeled learning, support, selection, convergence, restart and leakage variation. NOI-informed later learning is a dependent immutable successor science/ensemble/uncertainty generation, not independent validation or a retroactive edit. Changing to per-realization relearning of model, support, cleaner, coefficients, stopping or selector requires a separate method/ensemble and complete member-state graph. Members under unchanged rules get new state/application generations, not a method each. Fixed/relearned members cannot share one ensemble or be pooled here.

Admitted source: RESPONSE_AND_UNCERTAINTY_FAMILIES; BOUNDARIES_AND_CONVENTIONS B4.

15 An iteration changes state under a declared method

S_0 is initial seed state. The generic transition records roles, not an update law:

$$(S_k, y_k^{\text{in}}) \longrightarrow (m_k^{\text{candidate}}, m_k^{\text{accepted}}, m_k^{\text{applied}}, \rho_k, z_k, S_{k+1}).$$

A nonoccurring role has a core-permitted **not_applicable** reason; a missing required fact has an **unavailable** cause. These are different from numerical zero. Bind immutable base parent, exact input, lineage, branch, absolute zero-based iteration index, method-specific identity/generation, parent generation, state generation and application generation.

Each method must say whether the input is the original parent, the previous result, a reconstruction from the original parent and cumulative model, or another exact product, including any needed domain conversion. Returning to the base and processing the prior result are different methods even with the same model update. The base parent remains in every lineage.

Rules and realized values have different identities

Changing a parent route, grouping, coefficient-recomputation rule, learning/model-construction rule, operator family, sign, role order, bypass graph, support rule, recurrence, stopping rule, terminal selector or failure behavior creates a new method identity. New coefficient values, learned states or candidate/accepted/applied models produced under unchanged rules instead have new state, iteration and application generations. They are not aliases.

Factor each operation and consequential state component into exactly one category: *fixed numerical operator*; *fixed structural state with recomputed coefficients*; *externally supplied*; *relearned*; *bypassed*; *not_applicable*; *unavailable*. Mixed stages must be factored. Each entry binds category, owner, parent/generation, dependencies, order, application count, path and failure. Different conditioning scopes have separate records. A generic fixed or bypass flag is insufficient (Appendix H).

PTC owns its estimator and application rule. Fixed loading subspace with recomputed coefficients, frozen numerical-component subtraction and fully relearned PTC state are distinct procedures. Recomputed coefficients do not turn one unchanged rule into a new method on each iteration. Neither a fixed basis nor recomputation proves a complete linear F_k . The exact frozen quotation and rank restrictions are retained in Appendix K.

Use grouping-neutral **application_scope_resolved** and separate **learned_candidate**, **resolved_model_state**, **applied_model_state**, **realized_iteration_state**. Not every model is fitted. Requested policy, effective plan, resolved state and realized application remain distinct. One complete method grouping binds membership, axes, scope and shared dependence. Observation-local, coadd-level and joint versus independent learning do not commute by implication.

Admitted source: ITERATION_STATE_LIFECYCLE; FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD; BOUNDARIES_AND_CONVENTIONS B1-B2.

16 Continuing, stopping and selecting are different claims

After completion of the iteration with absolute zero-based index N , continuation begins at absolute index $N + 1$ from S_{N+1} . N is an index, not a count. Exact continuation requires all state capable of changing a future required result, under the same compatible future controls and declared numerical profile. It must yield the same scientific states and results as uninterrupted execution. A terminal signal array alone does not meet this criterion.

The continuation record binds base parent; S_{N+1} ; separate model identities, states and any cumulative model; learned/operator state; supports/masks; update, selection and convergence history; counters/resource state; stochastic generator/key/counter state where applicable; method/parameter versions; branch/generations; additive reference/gauge and null-space state; and every upstream/model parent. This list does not excuse an omitted consequential fact. Materialized, compact or lineage-reconstructible forms must preserve exact scientific state queries and reconstruction/failure contracts. Representation alone does not change state identity.

Restart creates a new lineage with ancestry to reused state/parents; it is not continuation. Relearning under an unchanged rule makes a successor state/application generation and may belong to exact continuation. A rule change makes a new method. Seed, branch, retry, completed iteration, selected terminal and publication retain separate identities. A retry records attempt, state/randomness and failure accounting; it creates no observation or implicit reset (Appendix H).

A limit reached does not establish convergence

Keep `iteration_completed`, `resource_limit_reached`, `converged`, `stopped_by_policy`, `failed`, `selected_terminal` and `consumer_fit` distinct. A convergence claim binds proposition, quantity/domain, metric/norm, scale/denominator, tolerance, direction, persistence/consecutive rule, missing/non-finite behavior, generation and state. No value is selected here. Convergence establishes neither correctness, unit response, unbiasedness nor fitness.

Selection preserves the iteration it selects

A terminal product is a separate immutable selection record plus an immutable completed iteration bundle. The predeclared method-owned selection rule includes candidate population, ties, data access and failure behavior. Looking across iterations contributes to response and uncertainty; it cannot be hidden by reporting only the selected array.

The record has nine required fields: selected completed-iteration identity; complete candidate population with inclusion/exclusion states; rule/generation; convergence/stop/resource facts; causes; selected-terminal response; selected-terminal uncertainty; continuation/terminality state; downstream named-use status references. Appendix I fixes their contents. It never mutates, relabels or replaces the selected bundle. Missing a hard selection dependency blocks `selected_terminal`; no last or favorable iteration is selected by default.

Admitted source: `ITERATION_STATE_LIFECYCLE`; `SCOPE_BRIEF` §§4,6.

17 Completion is a local claim with hard dependencies

Every completed iteration bundle retains the universal roles: immutable base parent and exact input; method/lineage/branch/absolute iteration identities; target and separate candidate/accepted/applied models with spaces, transformations and equality facts; projected removal contribution, residual, processed residual, rejoin contribution and result; update/successor state; path support/influence and information origin; response and uncertainty states; validity/causes; continuation state; completion/failure state; and provenance. Removal and rejoin each bind their full additive compatibility, conversions, reference/gauge/null-space and mode-origin consequences.

A method may add required roles or bind optional companions needed for a named result/use. It cannot waive the universal structure or the following dependencies:

1. A numerical result needs exact input, applied operand and all required path contributions available or exactly reconstructible on the claimed support, with compatible removal/rejoin arithmetic. Missing contributions are never zero, identity or successful bypass.
2. Exact continuation needs every future-consequential state and compatible profile; a result array is insufficient.
3. Numerical response/uncertainty needs its declared premises, domains, inputs and conditioning. An unavailable companion permits only a narrower claim allowed by *both* the core and exact method, retaining status and cause.
4. Terminal selection needs a separate immutable reference to a completed iteration satisfying core and method requirements.

An absent role needs core-permitted `not_applicable` and an explicit method-defined narrower composition. Unavailable is not absent. These generic meanings claim no actual numerical bundle or completed iteration in this draft.

ALIGN/AST own coordinate/event meaning; RTC/CAL own conditioning/calibration; PTC owns cleaning and coefficient families; MAP/JINC own their estimands, normalization, grouping and numerical admission; NOI owns ensemble/uncertainty methods. FRUIT owns internal feedback selection, update, transition, convergence, stopping and terminal selection. Other producers retain their exact product/transformation authority. Appendix J preserves the route-conditional quantities and producer restrictions.

Downstream owners retain Pointing, OOF, source inference, catalogue and other named-use admission. SCI-VAL may bind and evaluate an approved named-use policy; it does not author/execute recurrence, select models, advance iterations or choose terminal products. A missing downstream profile does not erase an otherwise permitted complete local bundle; completeness does not confer fitness.

Consumers receive the exact underlying MAP, JINC, FLT-FIXED, FLT-MATCHED, detector-time or other method-defined product role plus terminal-FRUIT lineage. FRUIT ancestry creates no fifth generic numerical parent class. Naming a role does not admit it. Publication is separate from scientific validity or readiness.

Admitted source: `ITERATION_STATE_LIFECYCLE`; `BOUNDARIES_AND_CONVENTIONS` B3–B5; `SCOPE_BRIEF` §§4–5.

A Conditional predictions: composition limits

All twenty IDs below retain their Stage A meaning. Every additive comparison requires ASM-001; linear predictions require ASM-003 where stated. Reference models and perturbations belong to the same applied-model space, not accepted models by implication. Missing premises mean the affected claim is unavailable. These are conditional deductions, not experiments or selected defaults.

SCI-FRUIT-PRED-001: Zero applied feedback

Premises and consequence. If $m_k^{\text{applied}} = 0$ and $\Pi_k(0) = B_k(0) = 0$ on otherwise identical compatible state, support, reference/gauge/null-space state, $z_k = F_k(y_k^{\text{in}})$. Zero candidate or accepted model does not imply zero applied model; coordinate zero or an unavailable mode is not an observed zero.

Derivation / discriminator. Substitute the applied zero into Equation 1. Offsets in non-zero-preserving maps defeat this reduction even for zero applied values.

SCI-FRUIT-PRED-002: Identity residual procedure

Premises and consequence. If F_k is identity on compatible $Y_k = Z_k$, $\Pi_k = B_k$ on the applied operand, and support, reference/gauge/null-space arithmetic agree, $z_k = y_k^{\text{in}}$ in that exact space. No unavailable observational mode is restored and no accepted/applied equality is assumed.

Derivation / discriminator. Removal and rejoin cancel as equal compatible contributions. A missing premise blocks cancellation rather than supplying an alignment or missing measurement.

SCI-FRUIT-PRED-003: Zero net linear model path

Premises and consequence. Under complete fixed linearity and $B_k = A_k \circ \Pi_k$ on M_k , $D_k = 0$ and $z_k = A_k y_k^{\text{in}}$, independent of applied model under those conditions.

Derivation / discriminator. The two maps in the definition of D_k are equal. State or rule changes are outside the fixed-operator premise.

SCI-FRUIT-PRED-004: Perfect applied operand

Premises and consequence. If $y_k^{\text{in}} = \Pi_k(m_{\text{ref}}^{\text{applied}})$, $m_k^{\text{applied}} = m_{\text{ref}}^{\text{applied}}$ and F_k is zero-preserving in the declared reference/gauge, $z_k = B_k(m_{\text{ref}}^{\text{applied}})$. A perfect accepted model alone is insufficient.

Derivation / discriminator. Compatible subtraction makes $\rho_k = 0$. Nonidentity rejoin proves neither true sky nor unit response; content in an unavailable additive mode remains `model_or_prior_supported`.

Admitted source: REQUIRED_CONDITIONAL_PREDICTIONS 001-004; RESPONSE_AND_UNCERTAINTY_FAMILIES.

B Conditional predictions: error, procedure and support

SCI-FRUIT-PRED-005: Applied-model error

Premises and consequence. Under complete fixed linearity, the same input/operators and $m_k^{\text{applied}} = m_{\text{ref}}^{\text{applied}} + \delta_m^{\text{applied}}$, the model-induced difference from the same procedure at $m_{\text{ref}}^{\text{applied}}$ is $D_k \delta_m^{\text{applied}}$. Its conditional mean is that bias component, not all sky bias or untransformed accepted error.

Derivation / discriminator. Subtract the two instances of Equation 3; the identical input term cancels. The difference belongs to the declared compatible result space.

SCI-FRUIT-PRED-006: Recomputed versus frozen PTC component

Premises and consequence. Inherited ordinary fixed-subspace/recomputed-coefficient application and the separately named frozen-component subtraction have different rules/responses. Incidental equality does not make them interchangeable. One unchanged recomputation method retains identity across new coefficient/application states.

Derivation / discriminator. The quoted coefficient rule acts on its current centered input; the frozen subtraction holds its removed numerical component fixed and has identity input derivative. The distinct procedures are explicit in Appendix K.

SCI-FRUIT-PRED-007: Model-supported content

Premises and consequence. If a method permits model-only output or a declared model contribution in an upstream unavailable additive mode, that content is `model_or_prior_supported`, not observed, recovered from data, acquired exposure or unit-response truth.

Derivation / discriminator. Its declared information source is the model/prior. Rejoin and a reference/gauge choice add no measurement authority; permission for model-only output is itself required.

SCI-FRUIT-PRED-008: Support or additive incompatibility

Premises and consequence. For unequal path support or incompatible reference/gauge/null-space facts, use a permitted support class only with full declared additive compatibility; otherwise the affected composition is unavailable.

Derivation / discriminator. A partial or incompatible map has no licensed sum on that domain. No implicit fill, renormalization, offset or gauge alignment repairs the missing premise.

Admitted source: `REQUIRED_CONDITIONAL_PREDICTIONS 005-008; SUPPORT_INFORMATION_ORIGIN; PTC_APPLICATION_REFERENCE.`

C Conditional predictions: response and uncertainty

SCI-FRUIT-PRED-009: Data-dependent procedure

Premises and consequence. When construction, acceptance, application or consequential state depends on data, RF-03 includes one iteration with prior state fixed; RF-04 propagates fixed-sequence dependence; RF-05 includes stop/resource/selection dependence; RF-07 composes the full FRUIT-only parent-domain query.

Derivation / discriminator. Each boundary encloses different causal operations. Omitting an enclosed dependence answers another query. These are neither RF-01/RF-02 aliases nor independent duplicates of their components.

SCI-FRUIT-PRED-010: Crossing a query boundary

Premises and consequence. If a query crosses threshold, support, branch, state or terminal-selection boundaries, a derivative may be unavailable; retain the typed discontinuity or transition. A finite difference needs its own convention and compatible reference/gauge/null-space comparison.

Derivation / discriminator. A fixed-branch derivative does not describe a changed procedure branch. If comparison or derivative existence fails, that requested response is unavailable.

SCI-FRUIT-PRED-011: Joint conditional covariance

Premises and consequence. Under complete fixed linearity and the exact joint covariance/conditioning premises for $(y_k^{\text{in}}, m_k^{\text{applied}})$, all four terms of Equation 6, including both cross terms, are required unless exact authority establishes an omission. Candidate/accepted covariance needs transformation.

Derivation / discriminator. Expanding the covariance of the linear sum yields both mixed products. Conditioning on a fixed gauge cannot turn an unavailable mode into a measured zero-uncertainty mode or discard dependence from another target.

SCI-FRUIT-PRED-012: Reused information

Premises and consequence. Iterations reusing a parent or shared model/reference information do not become independent measurements and imply no $1/\sqrt{\text{iteration count}}$ uncertainty scaling.

Derivation / discriminator. Reusing information preserves ancestry and shared dependence. Iteration count supplies no independent-measurement population or scaling premise.

Admitted source: REQUIRED_CONDITIONAL_PREDICTIONS 009-012; RESPONSE_AND_UNCERTAINTY_FAMILIES.

D Conditional predictions: continuation and selection

SCI-FRUIT-PRED-013: Continuation equality

Premises and consequence. After completion of the iteration with absolute zero-based index N , if S_{N+1} is continuation-sufficient and future controls and method/profile agree, continuation begins at absolute index $N + 1$ from S_{N+1} and agrees with uninterrupted execution, including reference/gauge/null-space state. N is an index, not a count.

Derivation / discriminator. Both continuations consume the same complete scientific state under the same future controls. An omitted consequential fact defeats the premise. This criterion performs no replay or numerical sufficiency check.

SCI-FRUIT-PRED-014: Restart and relearning

Premises and consequence. Restart has a new lineage. Relearning under unchanged rules produces distinct state/application generations and may belong to exact continuation. Changing the learning rule requires a new method identity.

Derivation / discriminator. Lineage records the new initialization; state generation records an unchanged rule's new realization. Neither identity aliases the method's defining rules.

SCI-FRUIT-PRED-015: Resource limit

Premises and consequence. Reaching an iteration/resource maximum establishes its resource/stop state, not convergence or adequacy.

Derivation / discriminator. Resource accounting and the declared convergence proposition are separate facts. Exhaustion alone supplies none of the metric, persistence or adequacy premises.

SCI-FRUIT-PRED-016: Terminal identity

Premises and consequence. If the rule selects the completed iteration of absolute zero-based index N , the separate immutable record retains selected identity, complete candidate population, rule/generation, convergence/stop/resource facts, causes, terminal response/uncertainty, continuation/terminality and use-status references or permitted unavailable statuses.

Derivation / discriminator. Selection is a relation to an already completed bundle. Mutating, relabeling or replacing that bundle destroys the distinction required by the premise. Missing hard selection dependencies block the claim.

Admitted source: REQUIRED_CONDITIONAL_PREDICTIONS 013-016; ITERATION_STATE_LIFECYCLE.

E Conditional predictions: grouping and authority

SCI-FRUIT-PRED-017: Grouping does not commute by implication

Premises and consequence. Observation-local, coadd-level and jointly learned groupings do not imply $\text{FRUIT}(\text{Coadd}(y_o)) = \text{Coadd}(\text{FRUIT}(y_o))$. Any commutation needs exact compatible operators, learning and support premises.

Derivation / discriminator. Different grouping may change shared learning, state, support and dependence. This states absence of an implied equivalence, not universal inequality or a numerical coaddition rule. Here y_o denotes only the explicitly compared observation inputs of this conditional statement.

SCI-FRUIT-PRED-018: Distinct NOI rules

Premises and consequence. Fixed-state NOI and per-realization relearning rules require separate method/ensemble identities and cannot share one ensemble here. Members under an unchanged rule have distinct realized states, not a new method each.

Derivation / discriminator. Changing the rule changes which variations the ensemble includes. New values under one rule change realization identity. No mixing/pooling authority is supplied.

SCI-FRUIT-PRED-019: No numerical admission from a family

Premises and consequence. Without a complete exact owner-approved `FRUIT-FEEDBACK-METHOD/<method-name>@<revision>` record, no numerical FRUIT route or result is admitted. Creating the v0.1 family or authoring/freezing the conditional core does not approve a method-specific identity.

Derivation / discriminator. Authority C is independent of A/B and of document completion. Even an approved method still needs its required upstream numerical authority. Current numerical status remains `unavailable_pending_separate_owner_approval`.

SCI-FRUIT-PRED-020: Local completeness and consumer fitness

Premises and consequence. A bundle satisfying universal minimum structure/hard dependencies and its exact method's requirements does not establish Pointing, OOF, source/catalogue admission or production fitness. An absent downstream profile does not erase permitted local completeness; a method cannot waive core roles to claim it.

Derivation / discriminator. The consumer owns its named-use admission. Local completion and downstream fitness are different propositions with separate evidence and authority. No actual numerical completed bundle is claimed in this draft.

Admitted source: `REQUIRED_CONDITIONAL_PREDICTIONS 017-020; FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD; BOUNDARIES_AND_CONVENTIONS`.

F Notation and source-local cover

Symbol or role	Exact meaning
$\Pi_k : M_k \rightarrow Y_k$	Projected removal contribution map; not an inferred inverse or numerical projector.
$\rho_k \in Y_k$	Compatible input minus projected applied model.
$F_k : Y_k \rightarrow Z_k$	Complete residual-processing procedure; $F_k(\rho_k)$ is the processed residual, not the rejoined result.
$B_k : M_k \rightarrow Z_k$	Model contribution map at the exact rejoin point.
S_k, S_{k+1}	Scientific state consumed/produced by iteration k ; S_0 is the seed.
$A_k : Y_k \rightarrow Z_k$	Exact fixed linear F_k , only under its established premises.
$D_k : M_k \rightarrow Z_k$	$B_k - A_k \circ \Pi_k$, only under complete fixed linearity of all three maps.
Ω_k^{FRUIT}	Exact conditioning: parents, operators, support, states, generations, reference/gauge/null-space facts and assumptions held fixed.
<code>fruit.update_k</code>	Separately typed causal increment, model change, diagnostic difference or unavailable role; never inferred from z_k .
<code>fruit.cumulative_model_k</code>	Present only if the method assigns cumulative meaning; distinct from an increment and the accepted/applied model roles.

New FRUIT notation does not use x or r for generic signal, residual or model: those symbols remain reserved for paired KID readout coordinates. Bare Q_p, N_p, C_p and unqualified NOI sign/member symbols are not adopted. F_k names the residual procedure, never carried feedback state.

The frozen PTC quotation uses source-local g for a detector group and t for a time sample. Its $x_{g,t}$ is called `PTC.centered_row` in new FRUIT prose; $\hat{A}_g$ is `PTC.loading_matrix`; its coefficients are `PTC.application_coefficients`; $W_{g,t}^{\text{app}}$, $G_{g,t}^{\text{app}}$ and Θ_g are respectively `PTC.application_influence`, `PTC.normal_matrix` and `PTC.resolved_group_state`. This cover does not rename KID readout coordinates or modify the frozen equations. PTC-local symbols are not FRUIT model or operator aliases.

Admitted source: NOTATION_AND_ROLE_TAXONOMY; BOUNDARIES_AND_CONVENTIONS B2.

G Premise register and availability

SCI-FRUIT-ASM-001: compatible arithmetic

All additive equalities, differences and cancellations require the full pairwise quantity/reference/gauge/null-space compatibility of Section 9, with state, support and conditioning fixed as declared. A missing premise blocks the affected claim; no offset, zero fill or reference alignment is inferred.

SCI-FRUIT-ASM-002: fixed residual linearity

The three-term expansion requires the complete $F_k = A_k$ to be exact fixed linear on its declared domain and state. A fixed basis or coefficient recomputation alone establishes neither linearity nor nonlinearity. Affine offsets must remain in the full procedure when this premise does not hold.

SCI-FRUIT-ASM-003: complete fixed linearity

A_k, Π_k, B_k must all be exact fixed linear on declared spaces, units, support, state and conditioning before defining D_k , its linear applied-model response, or using the four-term covariance identity.

SCI-FRUIT-ASM-004: joint conditional covariance

In addition to ASM-003, use compatible real vector representations, finite joint second moments and exact Ω_k^{FRUIT} , including reference/gauge/null-space coordinates, with applied-model axes and all cross dependence.

SCI-FRUIT-ASM-005: response comparison

The exact query convention, boundaries, baseline/perturbed domains, states, branches/support and reference/gauge/null-space comparison must be bound. Derivative existence is a further premise; a finite difference is separately typed with its own convention. No numerical step is chosen.

SCI-FRUIT-ASM-006: causal continuation

All future-consequential state and the same compatible future controls/profile are required for equality with uninterrupted execution. No replay is performed.

All six are conditional scientific premises, not measured facts or supplied numerical realizations. The complete method record, numerical parent, target representation, model construction, maps, recurrence, grouping, support, learning schedule, convergence threshold, resource cap, terminal selector, response, covariance/NOI method and downstream named-use profiles remain `unavailable_pending_separate_owner_approval`. No placeholder method instance is created.

This draft makes no implementation-conformity, recovered-sky-mode, response/covariance-fidelity, uncertainty-coverage, convergence-adequacy, Pointing/OOF/catalogue-fitness, observational-validation, performance, readiness or production-suitability/authorization claim. No implementation, experiment, replay, numerical qualification, production or Unity activity was authorized or performed. No online adaptive estimator is introduced.

Admitted source: SCOPE_BRIEF §§7–10; RESPONSE_AND_UNCERTAINTY_FAMILIES; REQUIRED_CONDITIONAL_PREDICTIONS.

H Operation and lifecycle records

Operation/state category	Required content
Fixed numerical operator	Complete exact map and fixed state; linearity, domain and support remain separate propositions.
Fixed structural state with recomputed coefficients	Fixed structure, recomputation rule, dependencies and realized application state; not a frozen numerical subtraction.
Externally supplied	Owner, exact parent/version/generation and compatibility; external does not mean independent.
Relearned	Learning rule, parents/population, dependence, generation and failure; included in full-procedure claims.
Bypassed	Exact operation absent from model path, removal/rejoin locations and counts on both paths; residual state separately classified.
not_applicable	Exact absent role and permitted reason; not an unknown value.
unavailable	Missing/incompatible required fact, cause, domain and blocked operation/claim.

Factor mixed stages. Each entry has exactly one category at its exact conditioning scope, with owner, parent/generation, inputs, order, application count, path and failure. Record separate scopes separately.

Lifecycle identity	Required meaning
Seed	S_0 , lineage, exact source/model parents and method declarations.
Continuation	Same lineage/branch, compatible method/profile, after index N at index $N + 1$ from sufficient S_{N+1} .
Restart	New lineage, explicit reused ancestry; initialization/counters follow the method.
Relearned successor	New learned state/application generation and dependencies under unchanged rules; simultaneous restart makes a new lineage.
Branch	Distinct branch with parent state and departure; no relabeling earlier iterations.
Retry	Distinct attempt with method-defined state/randomness/failure accounting.
Completed iteration	Immutable input/state, result/state, k and local completion/failure facts.
Selected terminal	Separate immutable selector record and immutable completed bundle reference.
Publication	Separate publication state; no convergence, consumer-fitness or validation inference.

Admitted source: ITERATION_STATE_LIFECYCLE.

I The complete terminal-selection record

Each of these nine fields retains identity, scope, generation and cause, with values or permitted typed unavailable statuses. Hard dependencies remain binding.

1. Selected completed-iteration identity

Exact immutable selected iteration and bundle reference. Missing this blocks selection.

2. Complete candidate iteration population considered

Every considered iteration identity and inclusion/exclusion state under the rule, plus population identity; never just favorable members.

3. Exact selection rule and generation

Predeclared method-owned rule, parameters, generation, data access and tie/failure behavior.

4. Convergence, stopping and resource-limit facts

Separate realized propositions and resource accounting used by selection.

5. Selection causes

Why selection or failure occurred, including unavailable facts and affected scope.

6. Selected-terminal response

Exact RF-05 identity/status, any RF-07 composition reference, and additive-reference/gauge/null-space consequences.

7. Selected-terminal uncertainty

Exact identity, conditioning, reference/gauge/null-space consequences, omitted sources and typed status.

8. Continuation or terminality state

Exact continuation-state reference or terminality proposition/status; no inferred ability to continue from a selected array.

9. Downstream named-use status references

Exact consumer-profile/status references or truthful unavailable states; no inferred consumer fitness.

The record never mutates, relabels or replaces the selected bundle. An unavailable hard selection fact blocks the selected-terminal claim; response/uncertainty/use unavailability permits only the narrower claim allowed by both core and method. The completed iteration remains immutable even when a later selection fails. Numerical path components may be materialized or exactly reconstructible, with required queries and failure behavior preserved.

Admitted source: ITERATION_STATE_LIFECYCLE; FRUIT_FEEDBACK_METHOD_REQUIRED_RECORD.

J Inherited producer limits

PTC retains its calibrated-input transformation, correlated-mode cleaning, coefficient families, fitted/resolved/applied state and validity. Total removed signal is input minus output, distinct from the fitted correlated component alone. Its transformed output is not an independent sky estimate. Fixed-state linear null modes, modes made invariant/unidentifiable by re-estimating centering, and affine loss from a discarded additive reference are distinct losses. FRUIT binds placement and application state; it does not redefine the cleaner.

Every selected parent route must supply exact quantity, unit, frame, response, beam/template convention, calibration, support, validity, lifecycle and provenance, with all domain conversions. No route or universal unit is selected. *If* the ordinary PTC-to-MAP route is selected, its convention remains calibrated- x -derived nonpolarimetric total-intensity-equivalent, top-of-atmosphere point-source-equivalent mJy/beam with originating fixed nominal beam/template and spectral/calibration lineage. This conditional example establishes neither Stokes I, true sky, a generic FRUIT target nor a universal input unit. A STOKES label does not supply authority. A normalized map is not automatically a deconvolved or unbiased sky model.

MAP and JINC retain estimands, normalization, grouping, coefficients, coordinate bindings, support, validity and numerical admission. Exact PTC output, coefficient/QC identity and compatible coordinate association are producer/consumer facts; row/time/label coincidence is not the same processed occurrence. A MAP-permitted coefficient family does not imply JINC permission. JINC is a signed normalized observation-map estimator. Its base authority does not inherit ordinary MAP coaddition or expose a base response, variance, formal-weight, covariance, empirical-noise or significance product. FRUIT's conditional obligations manufacture none of those products. Formal weight is not covariance or significance. Filtered/unfiltered quantities are not interchangeable; no inverse or provisional filter is supplied.

NOI owns the exact ensemble/uncertainty method. The admitted SCI-NOI-ODQ-110C boundary is not a freeze of the entire NOI package or a numerical FRUIT route. Fixed-state parity, dependent successor information and per-realization relearning remain distinct as in Section 14. Fixed-state/relearned ensemble mixing receives no authority here.

Inherited identities are SCI-PTC v0.1/r0.5, SCI-MAP v0.1/r0.7.1 and SCI-JINC v0.1/r0.3, with only the supplied excerpts and boundaries admitted. Their frozen descriptions do not supply absent coefficients, parameters, support/adequacy decisions or response information. Original files were not reopened. Any unresolved boundary conflict must identify its exact claim and owner; it cannot be resolved by silently modifying upstream science.

Admitted source: BOUNDARIES_AND_CONVENTIONS B2-B5.

K Frozen PTC application quotation

This is the byte-exact permitted SCI-PTC v0.1/r0.5 application fragment, under the source-local notation cover in Appendix F. It supplies no numerical FRUIT cleaning, rank, metric, model or parent route. FRUIT imports the distinction and restrictions, not a fresh PTC derivation.

For the ordinary configured-rank PCA/SVD route, application uses the frozen group-local detector-loading subspace and mask-aware linear coefficient recomputation. For centered row $x_{g,t} = Y_{g,t}^{\text{CAL}} - \lambda_g$,

$$\begin{aligned} G_{g,t}^{\text{app}} &:= \widehat{A}_g^T W_{g,t}^{\text{app}} \widehat{A}_g, & \text{rank}_{\tau_g}(G_{g,t}^{\text{app}}) &= k_{\text{req},g}, \\ \widehat{m}_{g,t}^{\text{app}} &= x_{g,t} W_{g,t}^{\text{app}} \widehat{A}_g (G_{g,t}^{\text{app}})^+, \\ z_{g,t} &= x_{g,t} - \widehat{m}_{g,t}^{\text{app}} \widehat{A}_g^T. \end{aligned} \tag{7}$$

The exact output support determines which entries of $z_{g,t}$ are published. The solve is authorized only when every required input and $G_{g,t}^{\text{app}}$ is finite and its numerical rank under frozen tolerance τ_g equals $k_{\text{req},g}$. Otherwise application and output for that group-time are unavailable with cause. The generalized inverse does not authorize partial-rank subtraction, rank reduction, interpolation, or borrowing detectors from another group. Detector binding, metric, support, and insufficient-support state are frozen in Θ_g . On complete uniform support this reduces to detector-right projection through $I - \widehat{A}_g(\widehat{A}_g^T \widehat{A}_g)^+ \widehat{A}_g^T$. Any temporal-left, two-sided, detector/time-specific, or general vectorized family remains separately named and is not inferred from matrix orientation or rank.

A separately named frozen-component subtraction family is

$$\widehat{U}_{\text{total}} := U_{\text{total},\Theta}(Y^{\text{CAL}}), \quad \mathcal{A}_{\widehat{U}_{\text{total}}}^{\text{sub}}(Y') = Y' - \widehat{U}_{\text{total}}, \quad D\mathcal{A}_{\widehat{U}_{\text{total}}}^{\text{sub}}[Y](H) = H. \tag{8}$$

It is not interchangeable with Equation 7 and is not the base-v0.1 PCA/SVD application rule.

Admitted source: PTC_APPLICATION_REFERENCE.tex, exactly preserved; BOUNDARIES_AND_CONVENTIONS B2.